Exam *Statistics for Computer Scientists* 2023-2024

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The exam consists of 8 questions on 10 pages. Mark each answer with the question number. For each question and sub-question, the maximum number of points to be obtained is indicated (grade = 1 + total number of points obtained).

The exam will begin at 9:00 am and end at 12:00 pm. Students entitled to extra time have until 12:30 pm. Participation in the exam requires attendance for at least 45 minutes.

The exam is open book; it is permitted to use 1) an original hard copy of 'Statistical methods for the social sciences'; and/or 2) a (graphing) calculator. Use of other sources is not permitted.

On the last page of the exam, you will find a copy of the probability tables relevant to this exam. After the exam, you may take this question sheet and any scrap paper with you.

Explain your answers where asked, and write down any formulas used where applicable. Be concise in your answers.

Good luck!

(2) 1. During Dutch national elections, an early indication of the voting results is obtained by means of an exit poll, i.e., a shadow election where people leaving the voting station are asked to vote again. These results are usually available as soon as voting stations close, and generally provide an accurate impression of the final election result. To obtain an accurate estimate, the sample of people included in the exit poll should be representative of the population (i.e., all voters).

The sampling strategy of the exit poll is described as follows: "An exit poll is a shadow election. We select a number of polling stations throughout the Netherlands. Our employees ask voters who leave the polling station if they want to 'vote' again. The voter then receives a ballot paper showing the same parties as on the official ballot paper. Voters can then anonymously deposit their ballot paper."

($\frac{1}{2}$) (a) What is the correct statistical term for this sampling strategy?

> **Solution.** Cluster sampling: polling stations are the clusters, and a number of clusters are selected to be the sample. In each polling station (cluster), all voters are asked to participate.

($\frac{1}{2}$) (b) Two types of bias could occur using the sampling strategy described above. One can be quantified immediately, the other will only become apparent when the results

from the exit poll are compared with the final election result. Name both types of bias and describe how you could quantify (if possible) or mitigate them.

> **Solution.** Sampling bias (undercoverage): the selected polling stations may not be representative. To mitigate this, selection of polling stations should be carefully made so it is a good cross-section of all polling stations in the Netherlands. We only know this once the final election results are known.
>
> Nonresponse bias: not all voters may want to participate in the exit poll, and we don't know if the voters that participate in the exit poll are representative of all voters in those polling stations. We can quantify this immediately by keeping track of how many people refuse to participate in the exit poll.

(1) (c) As with any sample survey, margins of error apply to the exit poll. In practice, the result of the exit poll always deviates from the actual result by one seat for several political parties. There are 150 seats to be distributed, assume for simplicity that each 0.67% of votes earns a party one seat. How large should the sample size be to estimate the election results to within one seat, with 99% probability? Assume that the largest party will obtain at most 40% of all votes.

> **Solution.** The desired margin of error is 0.0067. We use this formula: $n = \pi * (1-\pi)(\frac{z}{M})^2$. We use $\pi = 0.4$ (a safe estimate for the maximum proportion), $z = 2.575$ (corresponding to a 99% confidence level) and $M = 0.0067$.
>
> $n = 0.4 * 0.6 * (\frac{2.575}{0.0067})^2 = 35450$
>
> The sample size should be at least 35 450

(2) 2. The yearly Netflix Engagement Report contains viewing data on more than 18 000 shows. The following variables are available:

- Title: the title of the show
- Season: which season of the show (number)
- Global availability: whether the show is available globally (yes or no)
- Unique viewers: the number of unique viewers (number)
- Hours viewed: total number of hours this show was streamed (hours)
- Language: the original spoken language of the show (twenty options)

For each of the research questions below, provide: 1) the explanatory variable and response variable; 2) the measurement scale of each of these variables; 3) the test you would choose to answer the research question (also note, where relevant, whether you choose the paired or unpaired version); and finally, 4) the hypotheses. **Write your answers in the provided table at the end of the exam and hand it in with the rest of your answers.**

($\frac{1}{2}$) (a) Does the number of unique viewers decline between the first and second season?

Solution. Could be interval (assuming the difference between subsequent seasons is equal) or ratio (assuming the ratio is also equal, "season 8 is twice as far into the show as season 4"). Categorical seems unlikely for season, so ordinal would be wrong.

Explanatory variable: season, interval/ratio

Response variable: unique viewers, ratio

Test: paired two-sample t-test, which simplifies to a one-sample t-test for the difference in unique viewers of both seasons

Hypotheses: $H_0 : \mu_1 = \mu_2, H_a : \mu_2 < \mu_1$, or $H_0 : \mu_d = 0, H_a : \mu_d < 0$ where d indicates the difference in unique viewers of seasons 1 and 2.

($\frac{1}{2}$) (b) Is the proportion of shows that are globally available dependent on the show's language?

Solution. Explanatory variable: language, nominal

Response variable: global availability, nominal

(or the other way around)

Test: chi-squared test of independence

Hypotheses: H_0 : language and global availability are statistically independent; H_a : language and global availability are statistically dependent. Or H_0 : the proportion of shows that are globally available is the same for each language; H_a : the proportions are not the same.

($\frac{1}{2}$) (c) Is there an association between the total number of hours viewed and the number of unique viewers?

Solution. Explanatory variable: unique viewers, ratio

Response variable: hours viewed, ratio

(or the other way around)

Test: linear regression with a t-test for a slope, or Pearson's correlation test

Hypotheses: $H_0 : \beta = 0, H_a : \beta \neq 0$ or $H_0 : \rho = 0, H_a : \rho \neq 0$

($\frac{1}{2}$) (d) Do English shows on average get the same number of unique viewers as non-English shows?

Solution. Explanatory variable: language, nominal

Response variable: unique viewers, ratio

Test: t-test for difference in means in two groups, unpaired

Hypotheses: $H_0 : \mu_{english} = \mu_{non-english}, H_a : \mu_{english} \neq \mu_{non-english}$

($2\frac{1}{2}$) 3. Researchers have collected a dataset on penguins of three species living on the Palmer Archipelago in Antarctica. The following variables are available, for 344 penguins (no missing data): species (3 options: Adelie, Chinstrap, and Gentoo), island (3 options: Torgersen, Biscoe, and Dream), beak length (in mm), beak depth (in mm), flipper length (in mm), body mass (in grams), and sex (male or female). The aim of the research is to predict body mass, using (a subset of) the other variables. Using software to fit a multivariate linear regression model, the following results are obtained:

	Estimate	Std. Error	t	$P(> \lvert t \rvert)$
(intercept)	-5354.191	286.434	-18.693	< 0.001
beaklength	-3.185	4.891	-0.651	0.515
flipperlength	47.410	1.846	25.689	< 0.001
sex_is_male	33.604	41.642	8.083	< 0.001

Source	Sum of Squares	df
Regression	176298814	3
Residual	44427666	340
Total	220726480	343

($\frac{1}{2}$) (a) Write down the prediction equation (formally) and interpret the estimated parameters given above. Either use the given variable names, or clearly state what your notation means.

Solution. $\hat{y} = -5354.191 - 3.185x_1 + 47.410x_2 + 33.604x_3$, where x_1 is beak length, x_2 is flipper length, and $x_3 = 0$ if the penguin is female, $x_3 = 1$ if the penguin is male.

A hypothetical female penguin with a beak and flippers of 0 mm would have a predicted body mass equal to the intercept: -5354.191 grams. For each mm increase in beak length, keeping all other variables constant, the predicted body mass decreases by 3.185 grams. For each mm increase in flipper length, keeping all other variables constant, the predicted body mass increases by 47.410 grams. Predicted body mass for a male penguin is 33.605 grams higher than for a female penguin with the same beak and flipper length.

($\frac{1}{2}$) (b) Calculate and interpret R^2 of the given model and data.

Solution. $R^2 = \frac{TSS-SSE}{TSS} = \frac{220726480-44427666}{220726480} = 0.7987$. Interpretation: the prediction error decreases by (almost) 80% if we use the prediction equation to predict body mass, compared to using the sample mean to predict body mass.

(1) (c) Not quite satisfied with the obtained R^2, the researchers create a second model that includes species as additional predictor variable. They obtain the following results:

Source	Sum of Squares	df
Regression	189662962	5*
Residual	31063518	338
Total	220726480	343

**Note: species is a categorical variable with 3 options, so two dummy variables are needed in the model.*

Compare both models using the appropriate statistical test and decide if including species in the model significantly improves the model at $\alpha = 0.05$. Your solution should contain the name of the test, the calculation of a test statistic, a p-value range, and a conclusion (i.e., you do not have to give the assumptions and hypotheses).

Solution. We have two nested linear regression models, so we use an F-test to compare the models. Model 1 is the reduced model and model 2 is the complete model.
$F = \frac{(SSE_r - SSE_c)/df_1}{SSE_c/df_2}$ with $df_1 = 2$ (the number of extra parameters in the complete model) and $df_2 = n - (p+1) = 344 - (5+1) = 338$.
$F = \frac{(44427666 - 31063518)/2}{31063518/338} = 72.7$
At 2 and 338 degrees of freedom and $\alpha = 0.05$, the critical F-value is approximately 3.03. Our calculated F-value is much larger than the critical value at $\alpha = 0.05$, so we conclude that model 2 is a significantly better fit than model 1 with $p < 0.05$. Including species improves the predictions.

($\frac{1}{2}$) (d) The SSE of a model is a measure of how well the model fits the sample data. What is the corresponding measure that estimates how well the model predicts new, unseen observations? Do you expect it to be smaller or larger than the SSE?

Solution. PRESS is the predicted residual sum of squares. (To calculate PRESS, use leave-one-out cross-validation to predict y for each observation, using the model trained on the sample data excluding that observation.) As with all model fit indices, we expect it to be 'better' on the data the model was trained on, than on new data. Therefore, we expect the SSE to be smaller than the PRESS, as smaller values indicate better model fit for SSE and PRESS.

(1) 4. Someone claims that less than 50% of the Dutch population owns a cat. To check this statement, you ask a random sample of $n = 100$ people whether they own a cat, and conduct a one-sided z-test. You find that in your sample, 46 people own a cat.

($\frac{1}{2}$) (a) Perform the statistical test to see if you can verify or reject the claim based on your sample data, using $\alpha = 0.05$. Make sure to fully describe all steps.

Solution. Step 1 - Assumptions:

- The data are obtained using randomization
- The variable is categorical
- At least 10 observations per category

Step 2 - Hypotheses:
$H_0 : \pi = 0.5$, $H_a : \pi < 0.5$. Note that this is a one-sided alternative hypothesis.
Step 3 - Test statistic:
$z = \frac{\hat{\pi}-\pi_0}{se_0}$ with $se_0 = \sqrt{\frac{\pi_0(1-\pi_0)}{n}}$
$se_0 = \sqrt{\frac{0.5(1-0.5)}{100}} = 0.05$
$z = \frac{0.46-0.50}{0.05} = -0.8$
Step 4 - P-value
We use Table A to find the left-hand tail probability of $z = -0.08$, which is equal to the right-hand tail probability of $z = 0.8$. $P = 0.2119$
Step 5 - Conclusion
Our calculated P-value is larger than the pre-specified significance level. Therefore, we cannot reject the null hypothesis. With this data, we have not found evidence that the population proportion of people owning a cat is less than 50%.

($\frac{1}{2}$) (b) If performed correctly, you found that the null hypothesis would not be rejected based on this sample. Keeping α and the sample size the same, what would be the maximum sample proportion for which you would reject the null hypothesis?

Solution. The z-value associated with a right tail probability of 0.05 is equal to $z = 1.645$. We also know that $z = \frac{\hat{\pi}-\pi_0}{se_0}$ with $se_0 = \sqrt{\frac{\pi_0(1-\pi_0)}{n}} = \sqrt{\frac{0.5(1-0.5)}{100}} = 0.05$.
We reject H_0 when $z < -1.645$ (one-sided alternative hypothesis):

$$\frac{\hat{\pi} - \pi_0}{se_0} < -1.645$$

$$\frac{\hat{\pi} - 0.5}{0.05} < -1.645$$

$$\hat{\pi} < 0.42$$

We may reject the null hypothesis if we find a sample proportion below 0.42.

($\frac{1}{2}$) 5. The 'Billboard 200' chart lists the 200 most streamed albums each week. In the last week of December, the chart contained 36 Christmas-themed albums, 9 albums titled 'Greatest Hits', and 10 albums by Taylor Swift. Assume that there is no overlap between these three categories.

($\frac{1}{4}$) (a) What are the odds that any random album from this chart is Christmas-themed?

Solution. There are 36 Christmas-themed albums and $200 - 36 = 164$ non-Christmas-themed albums. The odds are therefore $36/164 = 0.22$ that a random album on the list is Christmas-themed.

($\frac{1}{4}$) (b) What is the odds ratio that any random album from this chart is by Taylor Swift, versus being a Greatest Hits album? Interpret this odds ratio.

Solution. Odds for Taylor Swift: $10/190 = 0.052$

Odds for Greatest Hits: $9/191 = 0.047$

Odds ratio: $0.052/0.047 = 1.12$. The odds of an album being by Taylor Swift are slightly higher than the odds of being a Greatest Hits album.

($\frac{3}{4}$) 6. A multiple linear regression model is fit on a dataset containing one quantitative response variable and 55 quantitative explanatory variables. There are 500 observations in the dataset, 350 observations are used to train the model and the remaining 150 are used to assess the model fit. Software reports that SSE on the train set is 0, and SSE on the test set is 5000.

($\frac{1}{2}$) (a) Which issue likely explains the difference in SSE on the train and test sets?

Solution. Overfitting: the model is not only describes the structure but also the noise that is present in the train set. This allows for perfect prediction on the train set, but results in decreased predictive performance (= higher error) on new data such as the test set.

($\frac{1}{4}$) (b) What would be the best way to solve this issue?

Solution. Our model is overfitted, so we need to simplify it. The number of explanatory variables is quite large, so we should check for multicollinearity between them and exclude some variables. Or: collect more data.

($\frac{1}{2}$) 7. A COVID-19 self-test can be regarded as a statistical test that rejects the null hypothesis "there is no COVID-19 infection" when the p-value is below a certain α. As with any test, Type I or Type II errors may occur. Which type of error is less desirable in this

context, from a population health perspective? Which significance level would be better to use to make the decision: $\alpha = 0.05$ or $\alpha = 0.001$?

Solution. A type I error means that the null hypothesis is true, but is falsely rejected. In this case, the test indicates COVID-19 infection while there is none. A type II error means that the null hypothesis is false, but it is not rejected. In this case, the test would not indicate COVID-19 infection even though one is present. Type II error is worse in this scenario. To make the probability of a type II error smaller, we should use a larger significance level, so $\alpha = 0.05$.

F Distribution: Critical Values of *F* (5% significance level)

df_1 / df_2	1	2	3	4	5	6	7	8	9	10	12	14	16	18	20
1	161.45	199.50	215.71	224.58	230.16	233.99	236.77	238.88	240.54	241.88	243.91	245.36	246.46	247.32	248.01
2	18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.38	19.40	19.41	19.42	19.43	19.44	19.45
3	10.13	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81	8.79	8.74	8.71	8.69	8.67	8.66
4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96	5.91	5.87	5.84	5.82	5.80
5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74	4.68	4.64	4.60	4.58	4.56
6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06	4.00	3.96	3.92	3.90	3.87
7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64	3.57	3.53	3.49	3.47	3.44
8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35	3.28	3.24	3.20	3.17	3.15
9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14	3.07	3.03	2.99	2.96	2.94
10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98	2.91	2.86	2.83	2.80	2.77
11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90	2.85	2.79	2.74	2.70	2.67	2.65
12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80	2.75	2.69	2.64	2.60	2.57	2.54
13	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71	2.67	2.60	2.55	2.51	2.48	2.46
14	4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65	2.60	2.53	2.48	2.44	2.41	2.39
15	4.54	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59	2.54	2.48	2.42	2.38	2.35	2.33
16	4.49	3.63	3.24	3.01	2.85	2.74	2.66	2.59	2.54	2.49	2.42	2.37	2.33	2.30	2.28
17	4.45	3.59	3.20	2.96	2.81	2.70	2.61	2.55	2.49	2.45	2.38	2.33	2.29	2.26	2.23
18	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51	2.46	2.41	2.34	2.29	2.25	2.22	2.19
19	4.38	3.52	3.13	2.90	2.74	2.63	2.54	2.48	2.42	2.38	2.31	2.26	2.21	2.18	2.16
20	4.35	3.49	3.10	2.87	2.71	2.60	2.51	2.45	2.39	2.35	2.28	2.22	2.18	2.15	2.12
21	4.32	3.47	3.07	2.84	2.68	2.57	2.49	2.42	2.37	2.32	2.25	2.20	2.16	2.12	2.10
22	4.30	3.44	3.05	2.82	2.66	2.55	2.46	2.40	2.34	2.30	2.23	2.17	2.13	2.10	2.07
23	4.28	3.42	3.03	2.80	2.64	2.53	2.44	2.37	2.32	2.27	2.20	2.15	2.11	2.08	2.05
24	4.26	3.40	3.01	2.78	2.62	2.51	2.42	2.36	2.30	2.25	2.18	2.13	2.09	2.05	2.03
25	4.24	3.39	2.99	2.76	2.60	2.49	2.40	2.34	2.28	2.24	2.16	2.11	2.07	2.04	2.01
26	4.22	3.37	2.98	2.74	2.59	2.47	2.39	2.32	2.27	2.22	2.15	2.09	2.05	2.02	1.99
27	4.21	3.35	2.96	2.73	2.57	2.46	2.37	2.31	2.25	2.20	2.13	2.08	2.04	2.00	1.97
28	4.20	3.34	2.95	2.71	2.56	2.45	2.36	2.29	2.24	2.19	2.12	2.06	2.02	1.99	1.96
29	4.18	3.33	2.93	2.70	2.55	2.43	2.35	2.28	2.22	2.18	2.10	2.05	2.01	1.97	1.94
30	4.17	3.32	2.92	2.69	2.53	2.42	2.33	2.27	2.21	2.16	2.09	2.04	1.99	1.96	1.93
35	4.12	3.27	2.87	2.64	2.49	2.37	2.29	2.22	2.16	2.11	2.04	1.99	1.94	1.91	1.88
40	4.08	3.23	2.84	2.61	2.45	2.34	2.25	2.18	2.12	2.08	2.00	1.95	1.90	1.87	1.84
50	4.03	3.18	2.79	2.56	2.40	2.29	2.20	2.13	2.07	2.03	1.95	1.89	1.85	1.81	1.78
60	4.00	3.15	2.76	2.53	2.37	2.25	2.17	2.10	2.04	1.99	1.92	1.86	1.82	1.78	1.75
70	3.98	3.13	2.74	2.50	2.35	2.23	2.14	2.07	2.02	1.97	1.89	1.84	1.79	1.75	1.72
80	3.96	3.11	2.72	2.49	2.33	2.21	2.13	2.06	2.00	1.95	1.88	1.82	1.77	1.73	1.70
90	3.95	3.10	2.71	2.47	2.32	2.20	2.11	2.04	1.99	1.94	1.86	1.80	1.76	1.72	1.69
100	3.94	3.09	2.70	2.46	2.31	2.19	2.10	2.03	1.97	1.93	1.85	1.79	1.75	1.71	1.68
120	3.92	3.07	2.68	2.45	2.29	2.18	2.09	2.02	1.96	1.91	1.83	1.78	1.73	1.69	1.66
150	3.90	3.06	2.66	2.43	2.27	2.16	2.07	2.00	1.94	1.89	1.82	1.76	1.71	1.67	1.64
200	3.89	3.04	2.65	2.42	2.26	2.14	2.06	1.98	1.93	1.88	1.80	1.74	1.69	1.66	1.62
250	3.88	3.03	2.64	2.41	2.25	2.13	2.05	1.98	1.92	1.87	1.79	1.73	1.68	1.65	1.61
300	3.87	3.03	2.63	2.40	2.24	2.13	2.04	1.97	1.91	1.86	1.78	1.72	1.68	1.64	1.61
400	3.86	3.02	2.63	2.39	2.24	2.12	2.03	1.96	1.90	1.85	1.78	1.72	1.67	1.63	1.60
500	3.86	3.01	2.62	2.39	2.23	2.12	2.03	1.96	1.90	1.85	1.77	1.71	1.66	1.62	1.59
600	3.86	3.01	2.62	2.39	2.23	2.11	2.02	1.95	1.90	1.85	1.77	1.71	1.66	1.62	1.59
750	3.85	3.01	2.62	2.38	2.23	2.11	2.02	1.95	1.89	1.84	1.77	1.70	1.66	1.62	1.58
1000	3.85	3.00	2.61	2.38	2.22	2.11	2.02	1.95	1.89	1.84	1.76	1.70	1.65	1.61	1.58

Hand in this table with the rest of your solutions.

Name:

Student number:

	2a	2b	2c	2d
Explanatory variable Measurement scale				
Response variable Measurement scale				
Significance test				
Hypotheses				